THREAT INTELLIGENCE BRIEFING

Types of Malware & Their Workflows

Demystifying malware workflows and threat lifecycles

ATTACK LIFECYCLE ANALYSIS

01 DEFINING THE THREAT

Malware Is Not Magic

What it is

Malicious software is highly structured code designed to achieve specific objectives.

Logical execution

Whether the goal is theft, destruction, or extortion, it follows a predictable, logical sequence.

Why it matters

Understanding this workflow is the key to detecting and stopping it.

Malware obeys the rules of the operating system like any other software. Learn the rules it follows, and you can spot when they are being abused.

02 THE FRAMEWORK

The Universal Attack Lifecycle

01

Delivery

Phishing, vulnerabilities, or drive-by downloads.

02

Execution

Code runs via user action or automated triggers.

03

Persistence

A foothold that survives system reboots.

04

Command & Control

Beaconing to attacker servers for instructions.

05

Actions on Objectives

Encryption, exfiltration, or destruction.

Families differ, but nearly all malware must navigate these five stages to succeed.

03 WORKFLOW

Ransomware

GOAL

Financial Extortion

01 **Infiltration**
Breaches network, often via RDP or malspam.

02 **Enumeration**
Silently maps local files, network drives, and cloud backups.

03 **Key Exchange**
Contacts C2 servers to generate unique encryption keys.

04 **Encryption**
Rapidly locks high-value data with military-grade algorithms.

05 **The Demand**
Drops ransom notes demanding cryptocurrency.

04 WORKFLOW

Remote Access Trojans

GOAL

Stealthy persistent backdoor

PHASE 1

Masquerade

Hides inside seemingly legitimate software, like a cracked game or PDF.

PHASE 2

The Drop

User runs the benign wrapper, silently dropping the payload.

PHASE 3

The Open Door

Alters registry keys for persistence, opens a reverse shell.

PHASE 4

Full Control

Remote access to files, screen capture, or further malware.

The user believes the program works normally, unaware a backdoor runs in the background.

0 5 WORKFLOW

Worms

GOAL

Autonomous Propagation

01 Exploitation

Identifies and exploits network protocol vulnerabilities.

02 Installation

Installs its payload without human interaction.

03 Reconnaissance

Scans the local subnet and the internet for vulnerable hosts.

04 Replication

Copies itself to new targets, creating an exponential infection cycle.

06 WORKFLOW

Spyware & Keyloggers

GOAL

Covert Data Collection

01

Covert Install

Arrives bundled with free software or via browser exploits.

02

API Hooking

Hooks OS APIs to intercept keystrokes, clipboard, and history.

03

Log Storage

Encrypts and stores stolen data locally in hidden files.

04

Exfiltration

Pushes logs to drop servers via HTTPS or DNS tunneling.

By hooking into core OS functions, spyware reads passwords as they are typed, before they are ever encrypted.

07 WORKFLOW

Fileless Malware

GOAL

Evade Disk Forensics

01 The Concept

Never writes a malicious executable to the hard drive.

02 Injection

Injects shellcode directly into system memory (RAM).

03 Living off the Land

Hijacks built-in trusted OS tools like PowerShell or WMI.

04 Execution

Runs purely in memory, leaving virtually no artifacts behind.

08 WORKFLOW

Rootkits

GOAL

Deep System Subversion

01

Privilege Escalation

Secures SYSTEM or Kernel-level (Ring 0) privileges.

02

Interception

Modifies the operating system kernel itself.

03

The Cloak

Intercepts system calls for files, processes, or network ports.

04

Result

Lies to the OS, filtering its own processes from scans.

If antivirus asks the OS what is running, the rootkit intercepts the question and removes itself from the answer.

Defense Strategy

Detection

We hunt for behavioral anomalies, not just signatures. We watch for the workflow steps in action.

Mitigation

We break the chain. Blocking C2 communication can stop ransomware before encryption begins.

Response

We isolate infected hosts immediately to prevent lateral movement and propagation.

Every stage of the attack lifecycle is an opportunity to detect, disrupt, and defend.

END OF BRIEFING

Questions & Answers

Thank you for your attention. Let us discuss how these workflows shape our defense.

STAY VIGILANT